- **Module-1 (8 hours)**
- **Power Supplies** –Block diagram,
- Half-wave rectifier,
- Full-wave rectifiers and filters,
- Voltage regulators,
- Output resistance and voltage regulation,
- Voltage multipliers.
- **Amplifiers** – Types of amplifiers.
- Gain, Input and output resistance, Frequency response.
- Bandwidth, Phase shift,
- Negative feedback, multi-stage amplifiers.

SUNIL .M
ASSISTANT PROFESSOR,
ELECTRONICS AND COMMUNICATION ENGG.
JAIN INSTITUTE OF TECHNOLOGY,
DAVANGERE

Power supplies

- The block diagram of a d.c. power supply is shown in Fig. 6.1. Since the mains input is at a relatively high voltage, a step-down transformer of appropriate turns ratio is used to convert this to a low voltage.
- The a.c. output from the transformer secondary is then rectified using conventional silicon rectifier diodes to produce an unsmoothed (sometimes referred to as pulsating d.c.) output.
- This is then smoothed and filtered before being applied to a circuit which will regulate (or stabilize) the output voltage so that it remains relatively constant in spite of variations in both load current and incoming mains voltage.
- Figure 6.2 shows how some of the electronic components that we have can be used in the realization of the block diagram in Fig. 6.1.
- The iron-cored step-down transformer feeds a rectifier arrangement (often based on a bridge circuit).
- The output of the rectifier is then applied to a high-value reservoir capacitor. This capacitor stores a considerable amount of charge and is being constantly topped-up by the rectifier arrangement.

  The capacitor also helps to smooth out the voltage pulses produced by the rectifier.
- Finally, a stabilizing circuit often based on a series transistor regulator and a zener diode voltage reference provides a constant output voltage.

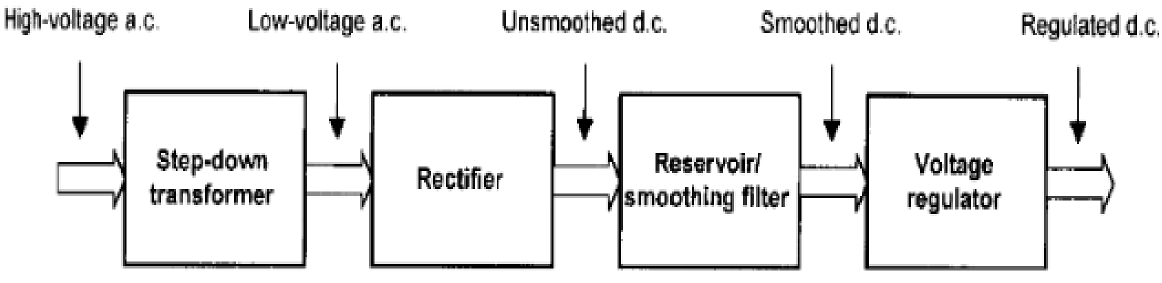

Figure 6.1 Block diagram of a d.c. power supply

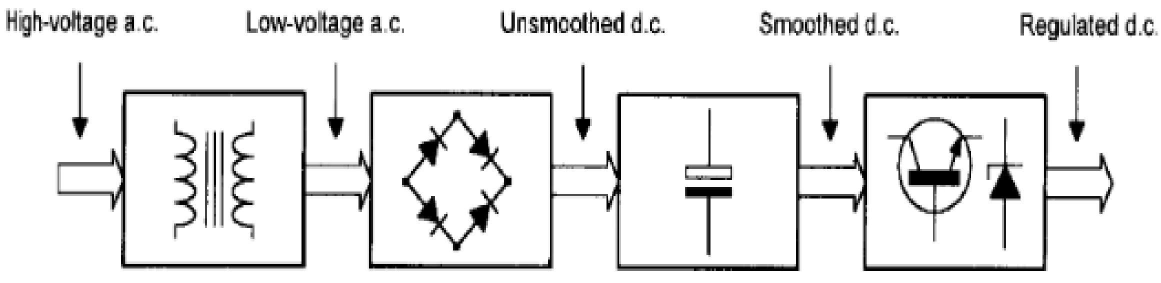

Figure 6.2 Block diagram of a d.c. power supply showing principal components

Rectifiers:

- Semiconductor diodes are commonly used to convert alternating current (a.c.) to direct current (d.c), in which case they are referred to as rectifiers.
- The simplest form of rectifier circuit makes use of a single diode and, since it operates on only either positive or negative half-cycles of the supply, it is known as a half-wave rectifier.

Half Wave Rectifier:

- Fig. 6.4 shows a simple half-wave rectifier circuit. Mains voltage (220 to 240 V) is applied to the primary of a step-down transformer (T1).
- The secondary of T1 steps down the 240 V r.m.s. to 12 V r.m.s. (the turns ratio of T1 will thus be 240/12 or 20:1).
- Diode D1 will only allow the current to flow in the direction shown (i.e.from cathode to anode). D1 will be forward biased during each positive half-cycle (relative to common) and will effectively behave like a closed switch.
- When the circuit current tries to flow in the opposite direction, the voltage bias across the diode will be reversed, causing the diode to act like an open switch (see Figs 1.2 (a) and (b), respectively).

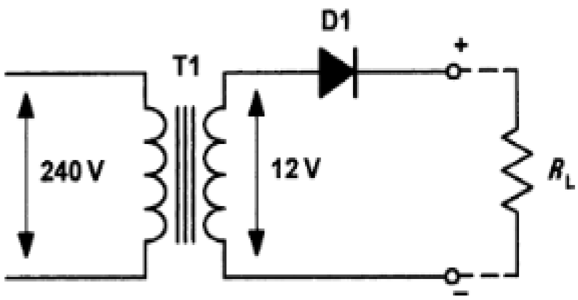

Fig 1.3 Half Wave Rectifier Circuit

- The switching action of D1 results in a pulsating output voltage which is developed across the load resistor (RL).
- During the positive half-cycle, the diode will drop the 0.6 V to 0.7 V forward threshold voltage normally associated with silicon diodes.
- However, during the negative half-cycle the peak a.c. voltage will be dropped across D1 when it is reverse biased. This is an important consideration when selecting a diode for a particular application.
- Assuming that the secondary of T1 provides 12 V r.m.s., the peak voltage output from the transformer's secondary winding will be given by:

$$V_{pk} = 1.414 \times V_{rms}$$

$$1.414 \times 12\ V$$

$$16.97V$$

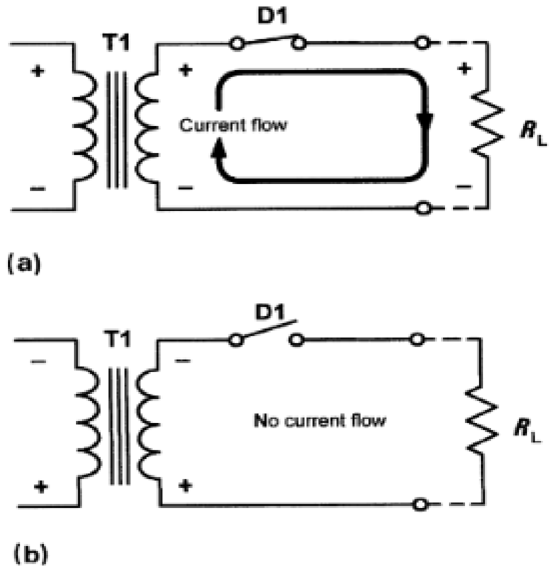

Fig 1.4 Switching action of Half Wave Rectifier Circuit.

- The peak voltage applied to D1 will thus be approximately 17 V.
- The negative half-cycles are blocked by D1 and thus only the positive half- cycles appear across RL.
- Note, however, that the actual peak voltage across RL will be the 17 V positive peak being supplied from the secondary on T1, minus the 0.7 V forward threshold voltage dropped by D1.
- In other words, positive half-cycle pulses having a peak amplitude of 16.3 V will appear across RL.

Reservoir and smoothing circuits:

Fig. 1.5 shows a considerable improvement to the circuit of Fig. 1.3.

- The capacitor, C1, has been added to ensure that the output voltage remains at, or near, the peak voltage even when the diode is not conducting.
- When the primary voltage is first applied to T1, the first positive half-cycle output from the secondary will charge C1 to the peak value seen across RL.
- Hence C1 charges to 16.3 V at the peak of the positive half-cycle. Because C1 and RL are in parallel, the voltage across RL will be the same as that across C1.
- Hence C1 charges very rapidly as soon as D1 starts to conduct. The time required for C1 to discharge is, in contrast, very much greater.
- The discharge time constant is determined by the capacitance value and the load resistance, RL.
- In practice, RL is very much larger than the resistance of the secondary circuit and hence C1 takes an appreciable time to discharge.
- During this time, D1 will be reverse biased and will thus be held in its non-conducting state.

- C1 is referred to as a **reservoir capacitor**. It stores charge during the positive half-cycles of secondary voltage and releases it during the negative half-cycles.

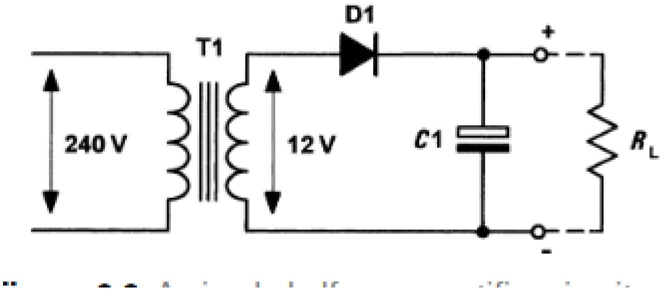

Fig 1.5A Simple Half Wave Rectifier Circuit with reservoir capacitor

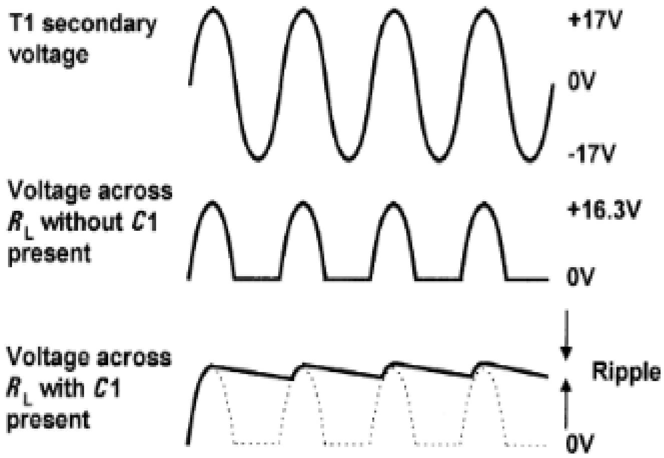

Fig 1.6 A Simple Half Wave Rectifier with reservoir circuit waveforms.

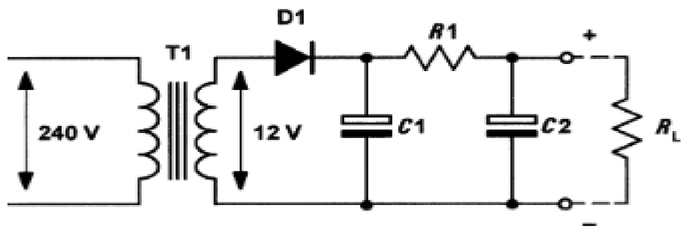

Fig 1.7 Half Wave Rectifier Circuit with R-C smoothing filter

- Fig. 1.6 shows the secondary voltage waveform together with the voltage developed across RL with and without C1 present. This gives rise to a small variation in the d.c. output voltage (known as ripple). Since ripple is undesirable we must take additional precautions to reduce it.

- One obvious method of reducing the amplitude of the ripple is that of simply increasing the discharge time constant. This can be achieved either by increasing the value of C1 or by increasing the resistance value of RL.
- Fig. 1.7 shows a further refinement of the simple power supply circuit. This circuit employs two additional components, R1 and C1, which act as a filter to remove the ripple.
- In effect, R1 and C1 act like a potential divider. The amount of ripple is reduced by an approximate factor equal to:

$$\frac{X_C}{\sqrt{R^2 + X_C{}^2}}$$

Problem:

The R–C smoothing filter in a 50 Hz mains operated half-wave rectifier circuit consists of R1 = 100 Ω and C 2 = 1,000 μF. If 1 V of ripple appears at the input of the circuit, determine theamount of ripple appearing at the output.

Solution

First we must determine the reactance of the capacitor, *C*1, at the ripple frequency (50 Hz):

$$X_C = \frac{1}{2\pi fC} = \frac{1}{6.28 \times 50 \times 1{,}000 \times 10^{-6}}$$

$$= \frac{1{,}000}{314} = 3.18\ \Omega$$

The amount of ripple at the output of the circuit (i.e. appearing across *C*1) will be given by:

$$V_{ripple} = 1 \times \frac{X_C}{\sqrt{R^2 + X_C{}^2}} = 1 \times \frac{3.18}{\sqrt{100^2 + 3.18^2}}$$

From which:

$$V = 0.032\ \text{V} = 32\ \text{mV}$$

Full-wave rectifiers

- Unfortunately, the half-wave rectifier circuit is relatively inefficient as conduction takes place only on alternate half-cycles.
- A better rectifier arrangement would make use of both positive and negative half-cycles.
- These full-wave rectifier circuits offer a considerable improvement over their half-wave counterparts.

- They are not only more efficient but are significantly less demanding in terms of the reservoir and smoothing components.

➢ There are two basic forms of full-wave rectifier;

1. The bi-phase type and
2. The bridge rectifier type.

Bi-phase rectifier circuits:

- Fig. 1.8 shows a simple bi-phase rectifier circuit.
- Mains voltage (240 V) is applied to the primary of the step-down transformer (T1) which has two identical secondary windings, each providing 12 V r.m.s. (the turns ratio of T1 will thus be 240/12 or 20:1 for each secondary winding).
- On positive half-cycles, point A will be positive with respect to point B. Similarly, point B will be positive with respect to point C.
- In this condition D1 will allow conduction (its anode will be positive with respect to its cathode) while D2 will not allow conduction (its anode will be negative with respect to its cathode). Thus D1 alone conducts on positive half-cycles.
- On negative half-cycles, point C will be positive with respect to point B. Similarly, point B will be positive with respect to point A.
- In this condition D2 will allow conduction (its anode will be positive with respect to its cathode) while D1 will not allow conduction (its anode will be negative with respect to its cathode). Thus D2 alone conducts on negative half-cycles.
- Fig. 1.9 shows the bi-phase rectifier circuit with the diodes replaced by switches.
- In Fig. 1.9 (a) D1 is shown conducting on a positive half-cycle while in Fig. 1.9 (b) D2 is shown conducting.
- The result is that current is routed through the load in the same direction on successive half-cycles.

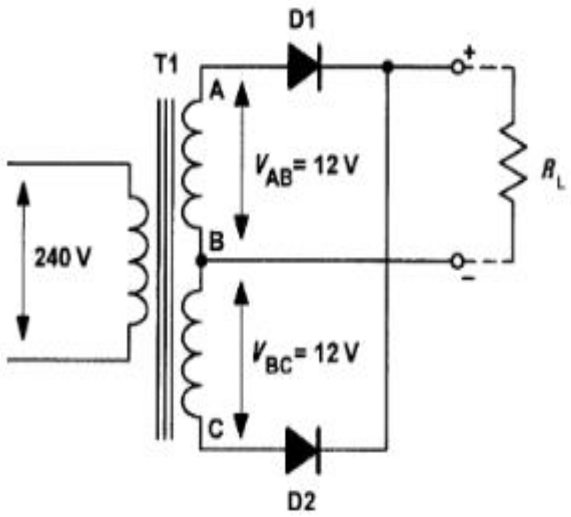

Fig 1.8 BI-PHASE Full Wave Rectifier Circuit

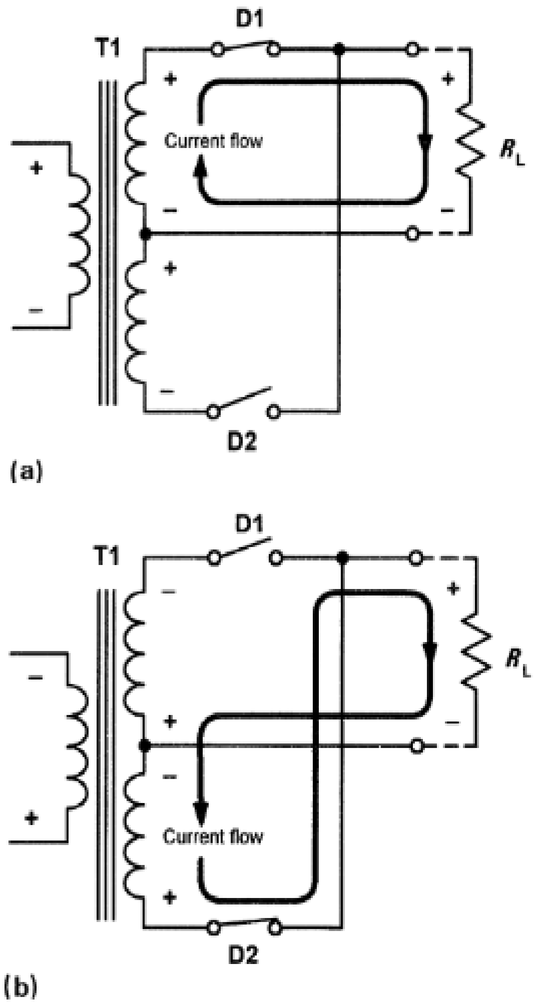

Fig 1.9 Switching action of BI-PHASE Full Wave Rectifier Circuit.

Reservoir and smoothing circuit of BI-PHASE full wave rectifier:

- Fig. 6.12 shows how a reservoir capacitor (C1) can be added to ensure that the output voltage remains at, or near, the peak voltage even when the diodes are not conducting.
- This component operates in exactly the same way as for the half-wave circuit, i.e. it charges to approximately 16.3 V at the peak of the positive half-cycle and holds the voltage at this level when the diodes are in their non-conducting states.
- C1 charges very rapidly as soon as either D1 or D2 starts to conduct.
- The time required for C1 to discharge is, in contrast, very much greater. The discharge time contrast is determined by the capacitance value and the load resistance, RL.
- In practice, RL is very much larger than the resistance of the secondary circuit and hence C1 takes an appreciable time to discharge.
- During this time, D1 and D2 will be reverse biased and held in a non-conducting state.
- Fig. 6.13 shows voltage waveforms for the bi-phase rectifier, with and without C1 present. Note that the ripple frequency (100 Hz) is twice that of the half-wave circuit.

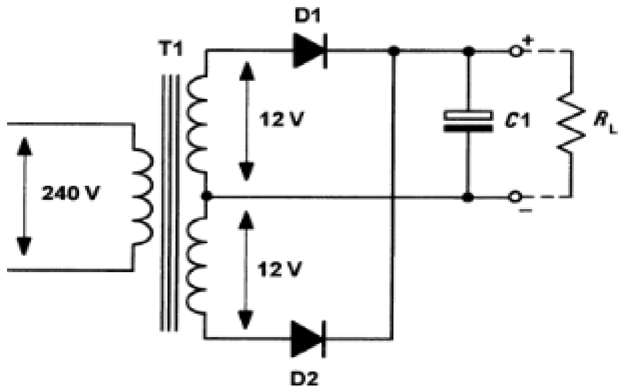

Fig 1.10 BI-PHASE Full Wave Rectifier Circuit with reservoir.

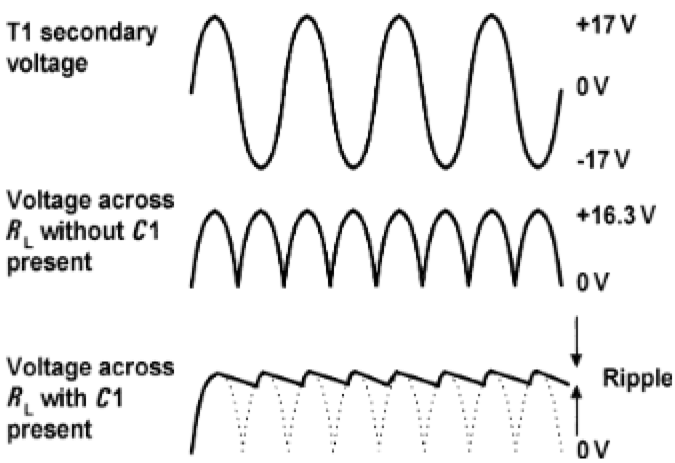

Fig 1.11 BI-PHASE Full Wave Rectifier Circuit with reservoir waveforms.

Bridge rectifier circuits:

- An alternative to the use of the bi-phase circuit is that of using a four-diode bridge rectifier in which opposite pairs of diodes conduct on alternate half-cycles.
- This arrangement avoids the need to have two separate secondary windings. A full-wave bridge rectifier arrangement is shown in Fig. 1.12.
- Mains voltage (240 V) is applied to the primary of a step-down transformer (T1). The secondary winding provides 12 V r.m.s. (approximately 17 V peak) and has a turns ratio of 20:1, as before.
- On positive half-cycles, point A will be positive with respect to point B. In this condition D1 and D2 will allow conduction while D3 and D4 will not allow conduction.
- Conversely, on negative half-cycles, point B will be positive with respect to point A. In this condition D3 and D4 will allow conduction while D1 and D2 will not allow conduction.
- Fig.1.13 shows the bridge rectifier circuit with the diodes replaced by four switches.
- In Fig 1.13(a) D1 and D2 are conducting on a positive half-cycle while in Fig. 1.13(b) D3 and D4 are conducting.
- Once again, the result is that current is routed through the load in the same direction on successive half-cycles.
- As with the bi-phase rectifier, the switching action of the two diodes results in a pulsating output voltage being developed across the load resistor (RL). Once again, the peak output voltage is approximately 16.3 V (i.e. 17 V less the 0.7 V forward threshold voltage).

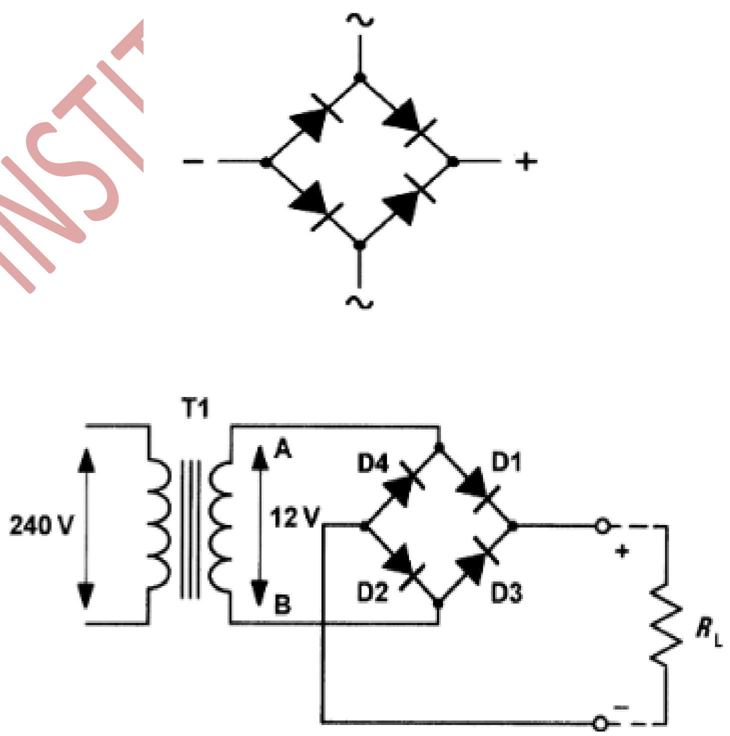

Fig 1.12 Full Wave Bridge Rectifier Circuit.

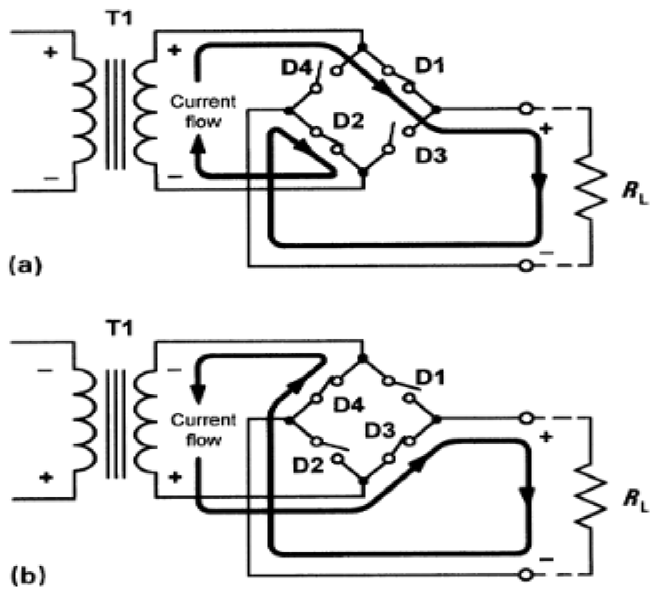

Fig 1.13 Switching action of Full Wave Bridge Rectifier Circuit.

Reservoir and smoothing circuit of full wave bridge rectifier:

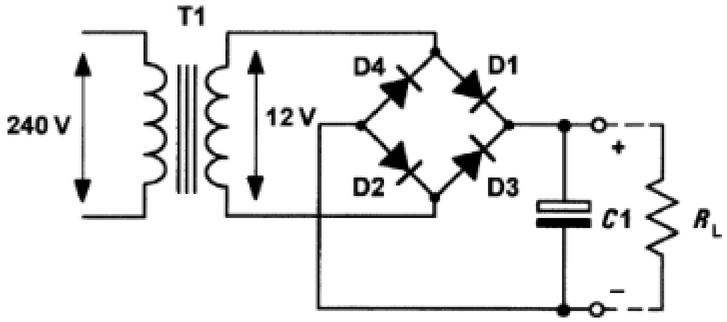

Fig 1.14 Full Wave bridge Rectifier Circuit with reservoir.

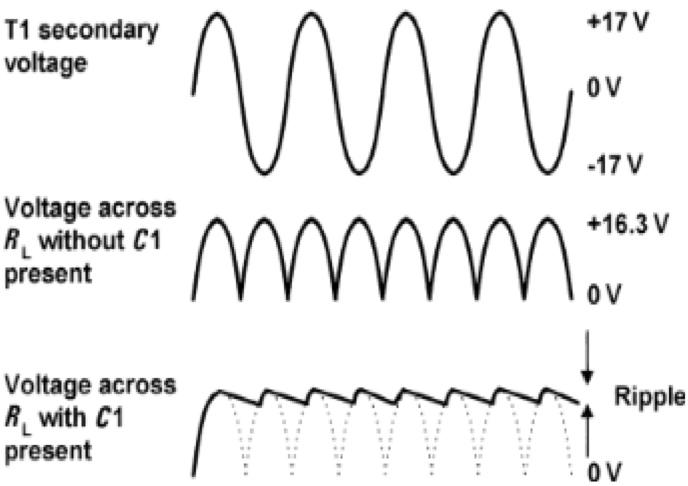

Fig 1.15 Full Wave Bridge Rectifier Circuit with reservoir waveforms.

- Fig. 1.14 shows how a reservoir capacitor (C1) can be added to maintain the output voltage when the diodes are not conducting.
- This component operates in exactly the same way as for the bi-phase circuit, i.e. it charges to approximately 16.3 V at the peak of the positive half-cycle and holds the voltage at this level when the diodes are in their non-conducting states.
- This component operates in exactly the same way as for the bi-phase circuit and the secondary and rectified output waveforms are shown in Fig. 1.15.
- Once again note that the ripple frequency is twice that of the incoming a.c. supply.
- Finally, R–C and L–C ripple filters can be added to bi-phase and bridge rectifier circuits in exactly the same way as those shown for the half-wave rectifier arrangement (see Figs 1.7).
- ## Voltage regulators

  A simple voltage regulator is shown in Fig. 1.16. RS is included to limit the zener current to a safe value when the load is disconnected.
- When a load (RL) is connected, the zener current (IZ) will fall as current is diverted into the load resistance (it is usual to allow a minimum current of 2 mA to 5 mA in order to ensure that the diode regulates).
- The output voltage (VZ) will remain at the zener voltage until regulation fails at the point at which the potential divider formed by RS and RL produces a lower output voltage that is less than VZ .

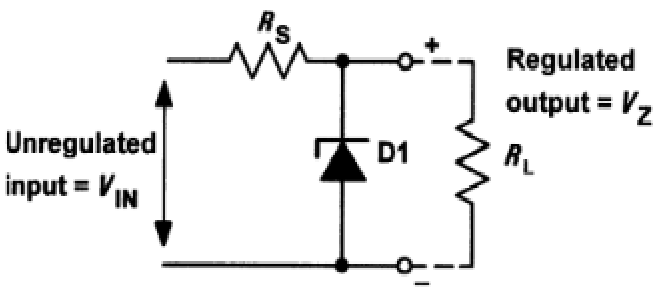

Fig 1.16 A simple zener voltage regulator.

The ratio of RS to RL is thus important. At the point at which the circuit just begins to fail to regulate:

$$V_Z = V_{IN} \times \frac{R_L}{R_L + R_S}$$

where V_{IN} is the unregulated input voltage. Thus the maximum value for RS can be calculated from:

$$R_S \text{ max.} = R_L \times \left(\frac{V_{IN}}{V_{IN}} - 1\right)$$

The power dissipated in the zener diode will be given by $P_Z = I_Z \times V_Z$, hence the minimum value for RS can be determined from the off-load condition when

$$R_S \text{ min.} = \frac{V_{IN} - V_Z}{I_Z} = \frac{V_{IN} - V_Z}{\left(\frac{P_Z \text{ max.}}{V_Z}\right)} = \frac{(V_{IN} - V_Z) \times V_Z}{P_Z \text{ max.}}$$

Thus:

$$R_S \text{ min.} = \frac{V_{IN}V_Z - V_Z^2}{P_Z \text{ max.}}$$

where P_Z max. is the maximum rated powerdissipation for the zener diode.

Problem:
A 5 V zener diode has a maximum rated power dissipation of 500 mW. If the diode is to be used in a simple regulator circuit to supply a regulated 5 V to a load having a resistance of 400 Ω, determine a suitable value of series resistor for operation in conjunction with a supply of 9 V.

Solution:
We shall use an arrangement similar to that shown in Fig. 1.15. First we should determine the maximum value for the series resistor, RS :

$$R_S \text{ max.} = R_L \times \left(\frac{V_{IN}}{V_{IN}} - 1\right)$$

thus:

$$R_S \text{ max.} = 400 \times \left(\frac{9}{5} - 1\right) = 400 \times (1.8 - 1) = 320\ \Omega$$

Now we need to determine the minimum value for the series resistor, R_S:

$$R_S \text{ min.} = \frac{V_{IN}V_Z - V_Z^2}{P_Z \text{ max.}}$$

thus:

$$R_S \text{ min.} = \frac{(9 \times 5) - 5^2}{0.5} = \frac{45 - 25}{0.5} = 40\ \Omega$$

Hence a suitable value for R_S would be 150 Ω (roughly mid-way between the two extremes).

Output resistance and voltage regulation:

- In a perfect power supply, the output voltage would remain constant regardless of the current taken by the load.
- In practice, however, the output voltage falls as the load current increases. To account for this fact, we say that the power supply has internal resistance (ideally this should be zero).
- This internal resistance appears at the output of the supply and is defined as the change in output voltage divided by the corresponding change in output current. Hence:

$$R_{out} = \frac{\text{change in output voltage}}{\text{change in output current}} = \frac{\Delta V_{out}}{\Delta I_{out}}$$

where ΔI_{out} represents a small change in output (load) current and ΔV_{out} represents a corresponding small change in output voltage. The regulation of a power supply is given by the relationship:

$$\text{Regulation} = \frac{\text{change in output voltage}}{\text{change in line (input) voltage}} \times 100\%$$

Problem:

The following data were obtained during a test carried out on a d.c. power supply:

(i) Load test

Output voltage (no-load) = 12 V

Output voltage (2 A load current) = 11.5 V

(ii) Regulation test

Output voltage (mains input, 220 V) = 12 V

Output voltage (mains input, 200 V) = 11.9 V

Determine (a) the equivalent output resistance of the power supply and (b) the regulation of the power supply.

Solution

The output resistance can be determined from the load test data:

$$R_{out} = \frac{\text{change in output voltage}}{\text{change in output current}} = \frac{12 - 11.5}{2 - 0} = 0.25\ \Omega$$

The regulation can be determined from the regulation test data:

$$\text{Regulation} = \frac{\text{change in output voltage}}{\text{change in line (input) voltage}} \times 100\%$$

thus

$$\text{Regulation} = \frac{12 - 1.9}{220 - 200} \times 100\% = \frac{0.1}{20} \times 100\% = 0.5\%$$

Voltage multipliers:

- By adding a second diode and capacitor, we can increase the output of the simple half-wave rectifier that we met earlier.
- A voltage doubler using this technique is shown in Fig. 1.17. In this arrangement C1 will charge to the positive peak secondary voltage while C2 will charge to the negative peak secondary voltage.
- Since the output is taken from C1 and C2 connected in series the resulting output voltage is twice that produced by one diode alone.

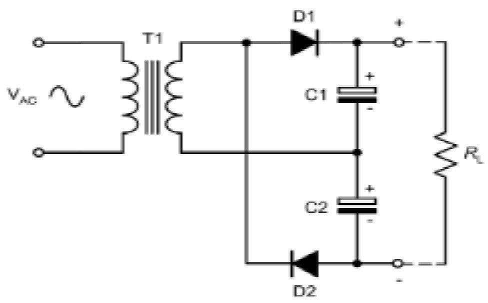

Fig 1.17 A voltage doubler.

- The voltage doubler can be extended to produce higher voltages using the cascade arrangement shown in Fig. 1.18.
- Here C1 charges to the positive peak secondary voltage, while C2 and C3 charge to twice the positive peak secondary voltage.
- The result is that the output voltage is the sum of the voltages across C1 and C3 which is three times the voltage that would be produced by a single diode.
- The efficiency of the circuit becomes increasingly impaired and high-order voltage multipliers of this type are only suitable for providing relatively small currents.

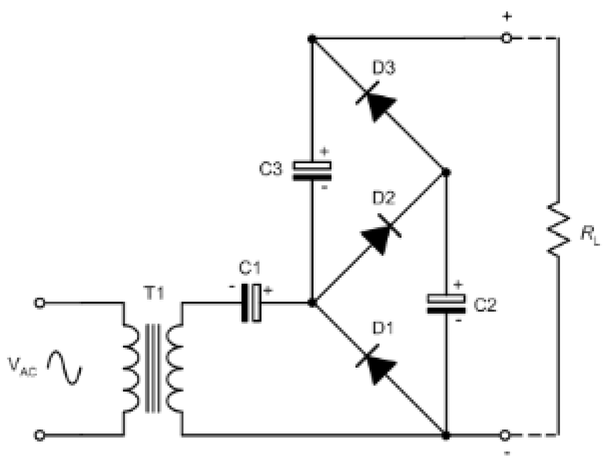

Fig 1.18 A voltage tripler.

Amplifiers:

- An amplifier is an electronic device that **increases the voltage, current, or power of a signal**.
- Amplifiers are used in wireless communications and broadcasting, and in audio equipment of all kinds.
- They can be categorized as either weak-signal amplifiers or power amplifiers

Types of amplifier

Many different types of amplifier are found in electronic circuits

a.c. coupled amplifiers

- In a.c. coupled amplifiers, stages are coupled together in such a way that d.c. levels are isolated and only the a.c. components of a signal are transferred from stage to stage.

d.c. coupled amplifiers

- In d.c. (or direct) coupled amplifiers, stages are coupled together in such a way that stages are not isolated to d.c. potentials. Both a.c. and d.c. signal components are transferred from stage to stage.

Large-signal amplifiers

- Large-signal amplifiers are designed to cater for appreciable voltage and/or current levels (typically from 1 V to 100 V or more).

Small-signal amplifiers

- Small-signal amplifiers are designed to cater for low-level signals (normally less than 1 V and often much smaller).
- Small-signal amplifiers have to be specially designed to combat the effects of noise.

Audio frequency amplifiers

- Audio frequency amplifiers operate in the band of frequencies that is normally associated with audio signals (e.g. 20 Hz to 20 kHz).

Wideband amplifiers

- Wideband amplifiers are capable of amplifying a very wide range of frequencies, typically from a few tens of hertz to several megahertz

Radio frequency amplifiers

- Radio frequency amplifiers operate in the band of frequencies that is normally associated with radio signals (e.g. from 100 kHz to over 1 GHz).

Low-noise amplifiers

- Low-noise amplifiers are designed so that they contribute negligible noise (signal disturbance) to the signal being amplified.
- These amplifiers are usually designed for use with very small signal levels (usually less than 10 mV or so).

Gain :

- One of the most important parameters of an amplifier is the amount of amplification or gain that it provides.
- Gain is simply the ratio of output voltage to input voltage, output current to input current, or output power to input power.
- These three ratios give, respectively, the voltage gain, current gain and power gain.

Thus:

$$\text{Voltage gain, } A_v = \frac{V_{out}}{V_{in}}$$

$$\text{Current gain, } A_i = \frac{I_{out}}{I_{in}}$$

$$\text{Power gain, } A_p = \frac{P_{out}}{P_{in}}$$

Note that, since power is the product of current and voltage (P = IV), we can refer that:

$$A_p = \frac{P_{out}}{P_{in}} = \frac{I_{out} \times V_{out}}{I_{in} \times V_{in}} = \frac{I_{out}}{I_{in}} \times \frac{V_{out}}{V_{in}} = A_i \times A_v$$

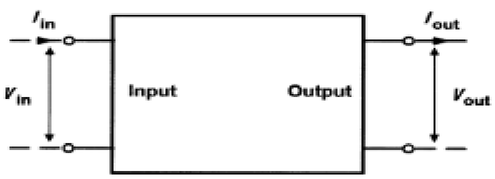

FIG.1.19. Block Diagram Of Amplifiers.

Problem:

An amplifier produces an output voltage of 2 V for an input of 50 mV. If the input and output currents in this condition are, respectively, 4 mA and 200 mA, determine:

(a) the voltage gain; (b) the current gain; (c) the power gain.

Solution

(a) The voltage gain is calculated from:

$$A_v = \frac{V_{out}}{V_{in}} = \frac{2\ \text{V}}{50\ \text{mV}} = 40$$

(b) The current gain is calculated from:

$$A_i = \frac{I_{out}}{I_{in}} = \frac{200\ \text{mA}}{4\ \text{mA}} = 50$$

(c) The power gain is calculated from:

$$A_p = \frac{I_{out} \times V_{out}}{I_{in} \times V_{in}} = \frac{200\ \text{mA} \times 2\ \text{V}}{4\ \text{mA} \times 50\ \text{mV}} = \frac{0.4\ \text{W}}{200\ \mu\text{W}} = 2{,}000$$

Note that the same result is obtained from:

$$A_p = A_i \times A_v = 50 \times 40 = 2{,}000$$

Input and output resistance:

- **Input resistance** is the ratio of input voltage to input current and it is expressed in ohms.
- The input of an amplifier is normally purely resistive (i.e. any reactive component is negligible) in the middle of its working frequency range (i.e. the mid-band).
- In some cases, the reactance of the input may become appreciable In such cases we would refer to input impedance rather than input resistance.
- **Output resistance** is the ratio of open-circuit output voltage to short-circuit output current and is measured in ohms.
- Note that this resistance is internal to the amplifier and should not be confused with the resistance of a load connected externally.
- As with input resistance, the output of an amplifier is normally purely resistive and we can safely ignore any reactive component.
- If this is not the case, we would once again need to refer to output impedance rather than output resistance.
- Fig. 7.8 shows how the input and output resistances are ‘seen’ looking into the input and output terminals, respectively.

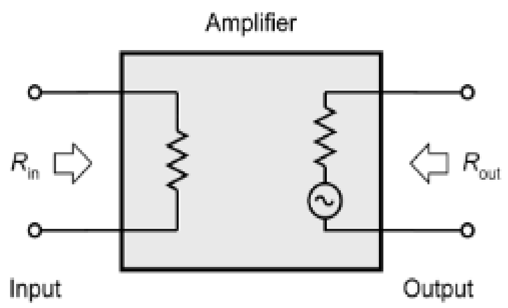

FIG.1.20. Input and output resistances 'seen' looking into the input and output terminals, respectively.

Frequency response

- The frequency response characteristics for various types of amplifier are shown in Fig. 1.21.
- Note that, for response curves of this type, frequency is almost invariably plotted on a logarithmic scale.
- The frequency response of an amplifier is usually specified in terms of the upper and lower cut-off frequencies of the amplifier.
- These frequencies are those at which the output power has dropped to 50% (otherwise known as the −3 dB points) or where the voltage gain has dropped to 70.7% of its mid-band value.
- Figs 1.22 and 1.23, respectively, show how the bandwidth can be expressed in terms of either power or voltage (the cut-off frequencies, f1 and f2, and bandwidth are identical).

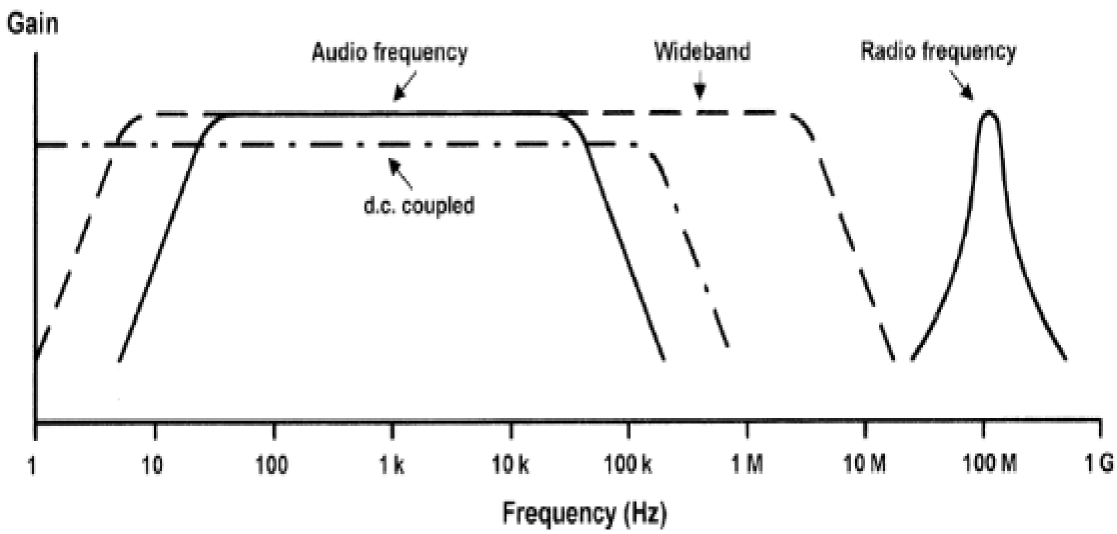

FIG.1.21. Frequency response and bandwidth

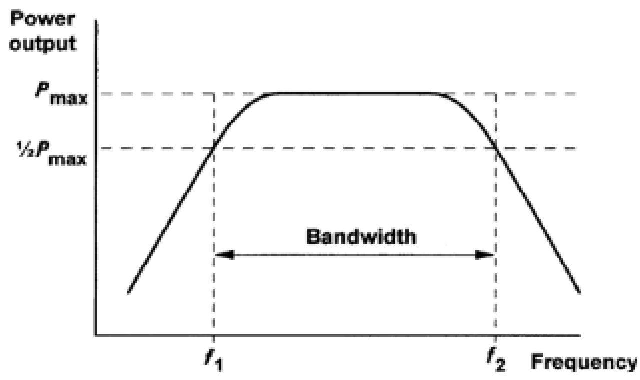

FIG.1.22. Frequency response and bandwidth (output power plotted against frequency)

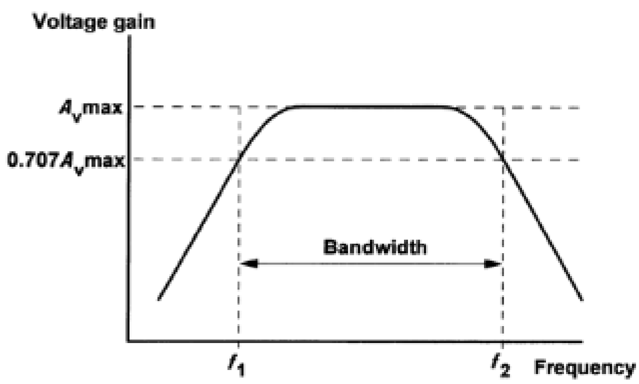

FIG.1.23. Frequency response and bandwidth (output voltage plotted against frequency)

Problem :

Determine the mid-band voltage gain and upper and lower cut-off frequencies for the amplifier whose frequency response is shown in Fig. 1.24.

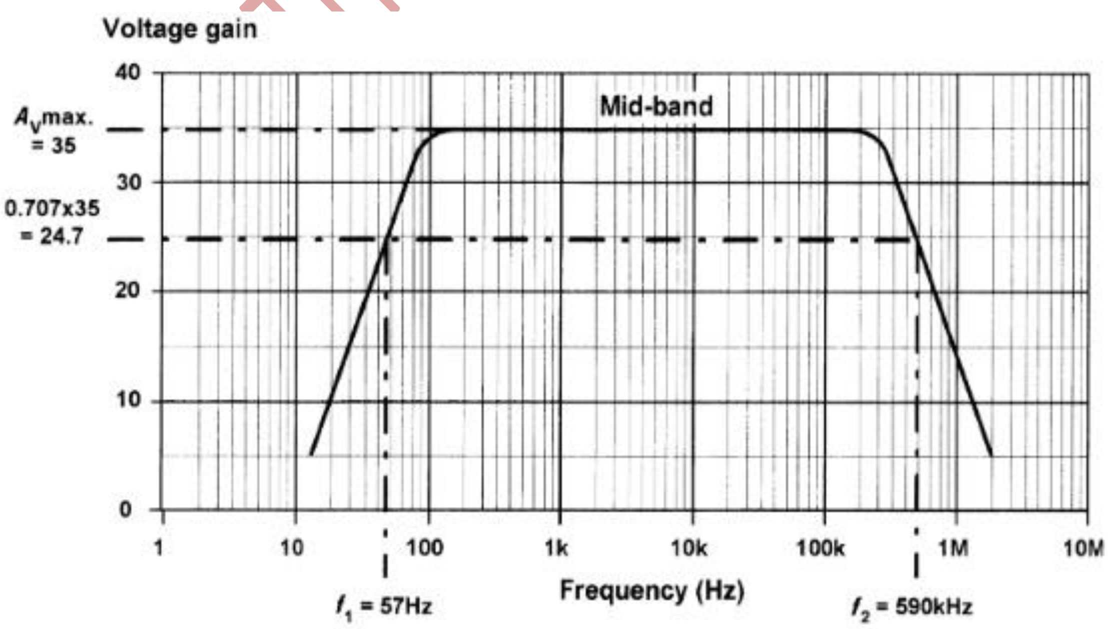

FIG.1.24. Frequency response and bandwidth for the above problem.

Solution

The mid-band voltage gain corresponds with the flat part of the frequency response characteristic. At that point the voltage gain reaches a maximum of 35 (see Fig. 1.24).

The voltage gain at the two cut-off frequencies can be calculated from:

Av cut-off = 0.707 × Av max

= 0.707 × 35 = 24.7

This value of gain intercepts the frequency response graph at f1 = 57 Hz and f2 = 590 kHz (see Fig. 1.24).

Bandwidth :

- The bandwidth of an amplifier is usually taken as the difference between the upper and lower cut-off frequencies (i.e. f2 − f1 in Figs 1.22 and 1.23).
- The bandwidth of an amplifier must be sufficient to accommodate the range of frequencies present within the signals that it is to be presented with.
- To reproduce a square wave, for example, requires an amplifier with a very wide bandwidth.
- Clearly it is not possible to perfectly reproduce such a wave, but it does explain why it can be desirable for an amplifier's bandwidth to greatly exceed the highest signal frequency that it is required to handle.

Phase shift:

- Phase shift is the phase angle between the input and output signal voltages measured in degrees.
- The measurement is usually carried out in the mid-band where, for most amplifiers, the phase shift remains relatively constant.
- Note also that conventional single-stage transistor amplifiers provide phase shifts of either 180° or 360°.

Negative feedback

- Many practical amplifiers use negative feedback in order to precisely control the gain, reduce distortion and improve bandwidth.
- The gain can be reduced to a manageable value by feeding back a small proportion of the output.
- The amount of feedback determines the overall gain.

- Because this form of feedback has the effect of reducing the overall gain of the circuit, this form of feedback is known as negative feedback.
- An alternative form of feedback, where the output is fed back in such a way as to reinforce the input (rather than to subtract from it) is known as positive feedback.
- This form of feedback is used in oscillator circuits.

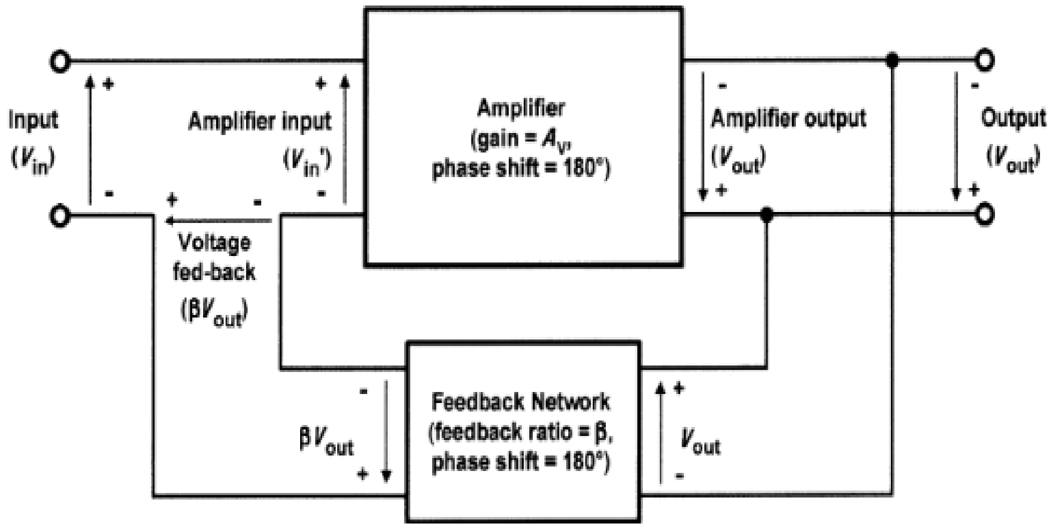

FIG.1.25 Amplifier with negative feedback applied.

Fig. 1.25 shows the block diagram of an amplifier stage with negative feedback applied.

In this circuit, the proportion of the output voltage fedback to the input is given by β and the overall voltage gain will be given by:

$$\text{Overall gain, } G = \frac{V_{out}}{V_{in}}$$

Now $V_{in}' = V_{in} - \beta V_{out}$ (note that the amplifier's input voltage has been reduced by applying negative feedback) thus:

$$V_{in} = V_{in}' + \beta V_{out}$$

and

$V_{out} = A_v \times V_{in}$ (note that Av is the internal gain of the amplifier)

Hence:

$$\text{Overall gain, } G = \frac{A_v \times V'_{in}}{V'_{in} + \beta V_{out}} = \frac{A_v \times V'_{in}}{V'_{in} + \beta(A_v \times V'_{in})}$$

Thus:

$$G = \frac{A_v}{1 + \beta A_v}$$

- Hence, the overall gain with negative feedback applied will be less than the gain without feedback.
- Furthermore, if Av is very large the overall gain with negative feedback applied will be given by: G = 1/β (when Av is very large) Note, also, that the loop gain of a feedback amplifier is defined as the product of β and Av.

Problem1:

An amplifier with negative feedback applied has an open-loop voltage gain of 50, and one-tenth of its output is fed back to the input (i.e. β = 0.1). Determine the overall voltage gain with negative feedback applied.

Solution

With negative feedback applied the overall voltage gain will be given by:

$$G = \frac{A_v}{1+\beta A_v} = \frac{50}{1+(0.1\times 50)} = \frac{50}{6} = 8.33$$

Problem2:

If, in problem1, the amplifier's open-loop voltage gain increases by 20%, determine the percentage increase in overall voltage gain.

Solution

The new value of voltage gain will be given by:

$A_v = A_v + 0.2A_v = 1.2 \times 50 = 60$

The overall voltage gain with negative feedback will then be:

$$G = \frac{A_v}{1+\beta A_v} = \frac{60}{1+(0.1\times 60)} = \frac{60}{7} = 7.14$$

The increase in overall voltage gain, expressed as a percentage, will thus be:

$$\frac{8.57-8.33}{8.33}\times 100\% = 2.88\%$$

Note that this example illustrates one of the important benefits of negative feedback in stabilizing the overall gain of an amplifier stage.

Problem3:

An integrated circuit that produces an open- loop gain of 100 is to be used as the basis of an amplifier stage having a precise voltage gain of 20. Determine the amount of feedback required.

Solution

Re-arranging the formula, $G = \frac{A_v}{1+\beta A_v}$ to make β the subject gives:

$$\beta = \frac{1}{G} - \frac{1}{A_v}$$

Thus:

$$\beta = \frac{1}{20} - \frac{1}{100} = 0.05 - 0.01 = 0.04$$

Multi-stage amplifiers :

- In order to provide sufficiently large values of gain, it is frequently necessary to use a number
  of interconnected stages within an amplifier.
- The overall gain of an amplifier with several stages (i.e. a multi-stage amplifier) is simply the product of the individual voltage gains.
  Hence:
  $$A_V = A_{V1} \times A_{V2} \times A_{V3}, \text{ etc.}$$
- Note, however, that the bandwidth of a multi-stage amplifier will be less than the bandwidth of each individual stage.
- In other words, an increase in gain can only be achieved at the expense of a reduction in bandwidth.
- Signals can be coupled between the individual stages of a multi-stage amplifier using one of a number of different methods shown in Fig. 1.26.

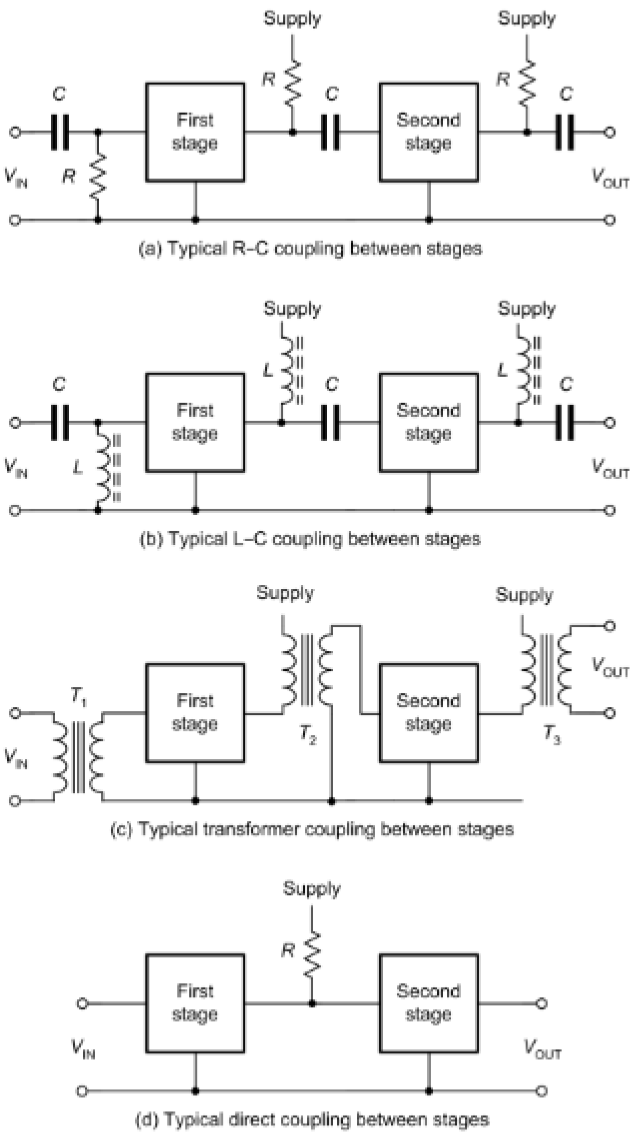

FIG.1.26 Different methods used for inter stage coupling

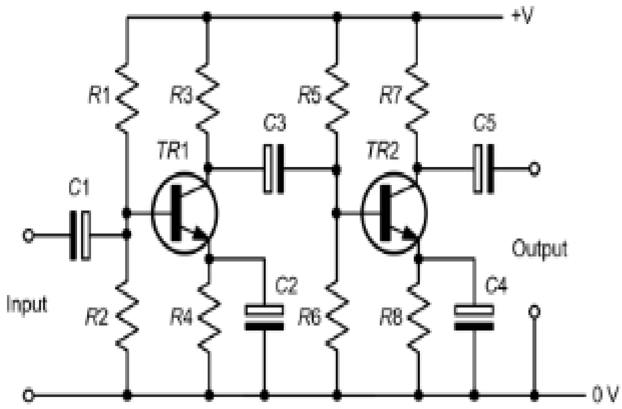

FIG.1.27 A typical two-stage high-gain R–C coupled common-emitter amplifier.

- The most commonly used method is that of R–C coupling as shown in In Fig. 1.26(a). In this coupling method, the stages are coupled together using capacitors having a low reactance at the signal frequency and resistors (which also provide a means of connecting the supply).
- Fig. 1.27 shows a practical example of this coupling method.
- A similar coupling method, known as L–C coupling, is shown in Fig. 1.26(b).
- In this method, the inductors have a high reactance at the signal frequency. This type of coupling is generally only used in RF and high-frequency amplifiers.
- Two further methods, transformer coupling and direct coupling, are shown in Figs 1.26(c) and 1.27(d), respectively. The latter method is used where d.c. levels present on signals must be preserved.